

Tampere University Unit of Computing Sciences
COMP.SE.610 Software Engineering Project 1
COMP.SE.620 Software Engineering Project 2

G05

Snowledge

Project plan

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Version history

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2.0	05/10/2025	Saku Hakamäki Eetu Koskela	Ch. 5.1.3 → Describe Sprint 1 outcome and Sprint 2 goals Ch. 6.2 → Define suggestions of metrics and acceptance levels Ch. 6.4 → Creating estimate for budget Ch. 7.1 → Add reworked risks and a new risk
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1 Introduction

1.1 Purpose and scope of project

This document is the project plan for the *Snowledge* group for the fall 2025 implementation of the Software Engineering Project course. This document outlines and defines the developmental goals for the project and serves as a source of information during the course. This document will be updated as the project evolves during the course if considered necessary.

The original scope for the project, as outlined in the customer's project proposal, focused on improving the existing application and introducing new features. However, this direction was revised due to input from an external consultancy, Mindhive, which the customer had employed to assess the current state of their existing application and recommended next steps.

Mindhive delivered two key documents, one detailing the state of the existing application and another presenting a high-level architectural vision and guide called the North Star. The assessment proved the old application to be quite limited in terms of scalability, maintainability, and overall quality, making it unsuitable for a system of its size and complexity. In contrast, The North Star proposed a reimagined, improved architecture, aligned with industry standards and best practises, to serve as instructions for developing a new application.

Based on this input from Mindhive, the customer proposed pursuing the complete rework rather than incremental improvements. The team agreed with this direction, and the project scope was defined accordingly, to follow the North Star vision, prioritize quality and develop the application as far as possible, making sure that future teams can continue its development without technical impediments.

During the Sprint 1 of the project, the project group was informed that another project group is also working for the customer but focusing on a different aspect. The other group was tasked to integrate the original existing implementation of the application to Suunto sports watches. However, it turned out that in order to complete this integration, the backend and database would need to be heavily altered. Since this task was already considered in the North Star document, the most reasonable solution to this conflict was seen to be that the two project groups join forces.

The high-level idea was that the other project group would take the responsibility in implementing the backend and database logic, as well as the Suunto watch integration. The remaining tasks for this project would be to implement the web and mobile applications. More detailed description can be found in Chapter 3.

1.2 Product and environment

The product is called Lumisovellus in Finnish and Snowledge in English. It consists of a web and mobile application for visitors and staff at Pallastunturi.

The application provides two main features:

1. Environmental monitoring – Users can view real-time data on snow levels, temperature, and wind conditions at Pallastunturi, as well as historical data from the past two days.

2. Safety assistance – Through the mobile application, users can request and receive help while in the Pallastunturi area. This feature is exclusive to the mobile version of Lumisovellus.

1.3 Customer's current system and other similar systems

The current version is in an operational state and is considered somewhat satisfactory by the customer. However, the project has been developed by various teams over the lifetime of the application. Due to this, technical debt has accumulated in the codebase and has resulted to a non-ideal system architecture. The system under development in this project is a rewrite of the existing one.

1.4 Project constraints

1.4.1 IPR

Customer has rights to commercially use and/or distribute the source code that is results from this project work. The group members retain rights for non-commercial use of the resulting work. If it happens so that part of the work, e.g. an algorithm, measurably improves on some well-known metric w.r.t. state-of-the-art as of 2025, the group members keep the rights to that part of the work.

1.4.2 GDPR

The application may collect certain personal information about the user such as:

1. Identifiers
 - a. First name, lastname
 - b. Email address
 - c. Phone Number
2. Location data
 - a. GPS-based precise location in Finland
 - b. Generalized geolocation (country-level)
3. Device and technical identifiers
 - a. IP address
 - b. Browser type / device type
 - c. Unique device identifiers (UDID, IDFA, etc.)
 - d. Mobile device brand/model, OS version
4. Communication data
 - a. Messages or reports that users send to support
 - b. Posts/content within the platform
 - c. Emails exchanged

As a result of this, the application should:

1. Explicitly map each processing activity to a lawful basis.
2. Implement a solid parental consent mechanism for children under applicable EU age limits.
3. Implement the ability for users to retrieve their data.
4. Implement the ability for users to delete their data.
5. Implement appropriate technical measures to protect their data.

1.5 Definitions, abbreviations and acronyms

Term	Definition
CI/CD	Continuous Integration and Continuous Delivery / Deployment. Set of automated software development practices that streamline the process of releasing new features and bug fixes.
CPU	Central Processing Unit. Computer component responsible for executing instructions in machine code.
Flutter	Framework for creating native apps for mobile, web, desktop, and embedded devices.
HSTS	HTTP Strict Transport Security. Web security standard that forces a web browser to always connect to a website using HTTPS.
HTTPS	Hypertext Transfer Protocol Secure. Encrypts exchanged data between web browser and server.
IDFA	Identifier for Advertisers. A unique random code assigned to a device by Apple. Allows advertising tracking and measurement.
Lumisovellus	Name of the application under development during this project.
Mindhive	Software development company, which has consulted the customer, and provided documentation of the current state and proposal of the next steps of the application.
MMT	Metrics Monitoring Tool
Monorepo	Software development strategy in which the code for multiple projects is stored in the same repository.
MySQL	Database management system
Next.js	Web development framework
North Star	Name of the document/instructions describing how the large-scale rewrite could be done for the existing application.
Pallaksen Pöllöt	Project customer. Ski school in Pallastunturi.
PM	Project Manager
PO	Product Owner
Snowledge	Name of the application under development during this project in English.
Suunto	Company making sports watches. Integrating the Snowledge app to Suunto watches is the responsibility of the other project group.
SW	Software
Trello	Kanban-style list-making application/service.
Turborepo	Build system for JavaScript and TypeScript codebases designed for monorepos.
UDID	Unique Device Identifier. A unique code assigned to a device, e.g. a mobile phone.
UI	User Interface. Here the human-computer interaction and communication happens.
WAF	Web Application Firewall. Filters, monitors and blocks malicious HTTP traffic to and from web applications.
w.r.t.	with respect to

2 Project organisation

2.1 Group members

Name	Role	Email	Skills
Saku Hakamäki	Project Manager	saku.hakamaki@tuni.fi	Software development, Documentation, Trello, Linux
Eero Kainulainen	Senior Developer	eero.kainulainen@tuni.fi	Full stack development, Databases, Pipelines, testing, Linux
Toivo Knuutinen	Senior Developer	toivo.knuutinen@tuni.fi	Full stack web development, databases & data model planning, Auth0, Linux
Kaapo Kontinen	Senior Developer	kaapo.kontinen@tuni.fi	SW development, Version Control, Linux, Technical communication, Data Visualization
Eetu Koskela	Product Owner	eetu.koskela@tuni.fi	SW testing (python, robot-framework), CI/CD (jenkins), SW development (web environments, python), Linux
Huyen Pham	Developer	hyuen.pham@tuni.fi	SW development (frontend & basic backend, database, APIs), version control, Linux
Eemeli Ruohomäki	Developer	eemeli.ruohomaki@tuni.fi	Software development (Web frontend and backend, Rust), UI/UX Design, Linux

Due to combining forces with the other project group, it was seen necessary to name a responsible person who would be aware of all things happening in both project groups. After discussions with the customer and other project group, the person designated for this role was Saku Hakamäki.

Each person of the project commits on working, on average, the number of hours necessary to meet the total requirement of 110-130 hours. These hours may be distributed across the duration of the project in a manner that ensures coverage of the weekly meetings as well as the work required each week.

2.2 Customer

Project customer is ski school [Pallaksen Pöllöt](https://pallaksenpollot.com).

2.2.1 Contact personnel

Name	Email
Juuso Holstein	juuso@pallaksenpollot.com
Arto Hämäläinen	artoahamalainen@gmail.com

2.3 Related organisations

2.3.1 Mindhive

Additional technical support is available during three scheduled meetings with a [Mindhive](#) contact, as arranged between the customer and them.

Name	Email
Ville Venäläinen	ville@mindhive.com

2.3.2 Tampere University

Name	Email	Role
Tero Ahtee	tero.ahtee@tuni.fi	Course staff
Pekka Mäkiahö	pekka.makiahö@istekki.fi / pekka.makiahö@tuni.fi	Project coach
Timo Poranen	timo.poranen@tuni.fi	Course staff
Kari Systä	kari.systa@tuni.fi	Course staff

2.3.3 Second Project Group

The table below highlights the members of the other project group working for the customer. This group has been assigned with the task of working on the backend and database, as well as integrating the Snowledge application to Suunto sports watches. One can also use the name G26 when referring to this group.

Name	Role	Email
Ibtehaj Abbas	Senior Developer	ibtehaj.abbas@tuni.fi
Nauman Afzal	Product Owner	nauman.afzal@tuni.fi
Waqas Hameed	Senior Developer	waqas.hameed@tuni.fi
Irteza Hammad	Senior Developer	irteza.hammad@tuni.fi
Wasay Rizwani	Project Manager	wasay.rizwani@tuni.fi
Sumair UI Hasan Wahab	Senior Developer	sumair.ulhasanwahab@tuni.fi

3 Project goals and ending/termination

3.1 High level goals of the project group

The primary goal of the project group is to implement as much of the “North Star” re-write as possible and deliver a working prototype that will eventually replace the current

multi-frontend, dual-backend setup. A secondary goal is to stay on the planned schedule within reasonable working hours (110 – 130 hours total) whilst completing course requirements in a way that achieves a final grade of 4 or higher. Finally, as a tertiary goal, the group aims to maximizing results while minimizing wasted work by working early and often, utilizing pair learning where possible and communicate actively.

3.2 High level goals of the customer

For the customer, the primary goal is that the app provides a good user experience, responsive performance that follows the "North Star" Rewrite. The second goal is to ensure the system uses a single backend with a strict, versioned REST API and a single database (MySQL with tracked migrations). Both client applications (Flutter app and Next.js web) must run on the same backend, with offline-first support and data caching for mobile. Finally, the customer also hopes to improve and demonstrate map functionality, including support for 3D maps.

3.3 High level goals and deliverable(s) of the project

The shared goal of the project is to develop a local version of the Lumisovellus "North Star" rewrite. This includes a single backend and MySQL database with migrations, a Flutter mobile client (offline-first) and a Next.js web client, a monorepo setup (TurboRepo) with pipelines, linting, and testing and a Docker based containerized environment for local development. The deliverables would be a working prototype with the core features mentioned above, along with documentation of the architecture and setup.

The main high-level objective for this project group is to implement the web and mobile frontends. Another project group, see Chapter 2.3.3. for contact information, has taken the responsibility of designing and implementing the backend and database areas of the North Star vision. The other group has also been assigned with the task of integrating the application to the Suunto sports watches.

3.4 Quitting (termination) criteria of the project

The project will be terminated if the group faces overwhelming technical difficulties, for example despite multiple attempts the prototype is unable to run according to the "North Star" rewrite. Termination may also occur if serious resourcing issues arises, such as key members quitting, leaving the remaining students unable to continue the work properly. In either case, termination of the project will be decided by considering the opinion of all involved parties, being the project group, the course instructor and the customer.

3.5 Ending criteria of the project

The project will be considered successfully completed when a local prototype is delivered before 12.12.2025. Documentation such as the test report and the final report must also be submitted before the mentioned date. The customer will review and accept the prototype, and the project will formally end once the course timeline is completed. From the courses' side, the final meeting must be held, the minutes submitted, and all action points closed. At an individual level, each student needs to have at least

110 working hours, and their peer reviews submitted. The authority to end the project lies with the group, customer and course instructor.

4 Project and Process management

4.1 Methods and tools

This project utilizes various tools for different purposes. For communication, Telegram and Discord have been chosen as the internal tools. Telegram is the main communication channel, and it is used to communicate the most important and urgent matters. A separate Discord server has been created to host internal meetings and to divide discussion to different topics, e.g. one discussion channel for frontend and one for backend. Discord is also used to store some information for later use, such as important links to different documents utilized across the course. For communication between the group and customer, email has been chosen as the main method. Also, a Telegram channel has been created to ease the communication between the customer and individual group members. After joining forces with the second project group, an additional Telegram channel was created for communication between the two separate project groups.

The version control system for this project has been chosen to be Git. The existing implementation of the project code was hosted in a GitLab repository, but it has been moved to GitHub to bypass the member limitation of GitLab and to allow each of the group members to access the source code directly.

The GitHub repo created by the customer was forked to the PM's personal repository, to allow the group members to utilize GitHub Pro features available free-of-charge to GitHub student accounts, e.g. defining more strict rules for branches. In the end of the project, this fork of the repo will be merged back to the GitHub repository created by the customer. To reduce the complexity of integrating the work of the two different project groups, both groups will be working and developing features in the same repository.

For project planning purposes, a Trello workspace has been established. The workspace contains two different Kanban boards, one general-purpose and one for development. These Kanban boards also serve as a visual representation of the workflow and allow the workspace users to easily check the state of the project. Trello showcases the different states of individual tasks, which are defined as done (DoD) after they have been documented and passed the corresponding tests created for them.

In addition to Trello, the Metrics Monitoring Tool (MMT) is also used for project management. MMT is used to log working hours and for creating project weekly reports, which contain different statistics from each week, such as spent working hours and realised risks. Analysing for risks and reacting to them is a key factor for a successful project. Managing risks is a continuous process, and they need to be considered on time to prevent them from having a negative effect on the schedule and objectives of the project.

During the project, Trello and MMT share a common goal of change management. The requirements of the project can evolve, new issues may arise during development, or the overall objectives can change in terms of new features. To prevent sudden

changes or new features to slow down the development, these changes need to be quickly assessed and integrated to the workflow.

When demo versions of the project are ready, those will be made available to the customer by utilizing the existing domain. For example, <https://demo.lumisovellus.fi> could act as the demo website, if possible. Otherwise, IP and port combination serving the demo could be used and provided to the customer. The website would be made available by utilizing the constructed CI/CD pipeline. For providing the demo iOS and Android mobile applications, TestFlight [3] and Google Play Internal App Sharing [4] could be utilized, respectively. At the time of writing, assigning one or many responsible persons for setting up the demo versions was not deemed necessary, since the process of doing so would be at least semi-automatic. If the situation changes, the assignment can be done as needed.

The scope of the project includes multiple documents, that are used store and showcase the work done from different aspects. The Project Plan document is used to discuss the overall scope of the project. The purpose of the Test Plan document is to clarify what is going to get tested, how it going to get done, and by who. The Test Report document shows what was tested, how it was done, and what were the results. The Final Report documents showcase the overall outcome of the project.

4.2 Planned meetings

The project has three main types of meetings, each of which have their own schedule and purpose.

Internal Weekly Meetings are for the group members to discuss the events and tasks in the project and development process. These topics can be upcoming or from the past week. They are held every Monday from 17:00 to 18:00. Every group member should participate, although exceptions are allowed. Any group member may call an extra meeting, if those are needed.

Customer Weekly Meetings are held to allow the group to share information on the project progress to the client and discuss possible issues. These meetings can be also held to ask any questions between the parties. These meetings are scheduled to happen bi-weekly (see Sprint Review Meetings below) on Thursdays from 09:00 to 10:00. The participants include at least the Project Manager (PM), Product Owner (PO) and one customer representative, but the participation of other representatives (group or customer) brings additional value to the meeting. These and any extra meetings are mainly called by the PM, PO or any customer representative.

Sprint Review Meetings are for inspecting the progress made during the past Sprint. These meetings are also held bi-weekly on Thursdays from 09:00 to 10:00, alternating between the regular Customer weekly Meetings described above. The participants of these meetings include the group members, customer representatives, and the coach. These meetings can be called by the PM, PO, any customer representative or the coach, but the main responsibility is with the group representatives, i.e. PM or PO.

4.3 Learning and study plan

Since the project and its scope includes many new elements for the group members, ongoing learning is needed for continuous progress of the project. To get familiar with

the existing implementation of the product, the customer provided a demonstration of how the current state of the product works, and what different elements it has. With the help of this demonstration the group can more easily get acquainted with product and the source code.

The customer has had some previous collaboration with Mindhive, a company who has provided helpful documentation on the current state of the product and a proposal for a rework of the product. These documents, especially the North Star, serve as a guide on the project, providing vital information and direction for development.

With the help of previously mentioned demonstration and documents, the group will take a deeper dive to the current implementation of the product. More experienced group members guide other group members and provide instructions to e.g. reproduce the development environment. If any tool-specific instructions or other manuals are needed, Kaapo Kontinen has been assigned as the responsible person to find them.

5 Project iterations and timing

5.1 Iterations (sprints)

The project work is divided into separate iterations, often referred to as sprints. These sprints are two weeks long, and they act as stepping stones towards completing the defined scope of the project. Table 5.1 shows the high-level schedule of the project. However, predicting the content of each implementation sprint or the end-state of the product at this stage is challenging. The group will implement as much of the project as possible while building a solid base for other groups to work on in the future.

Table 5.1. Project schedule.

Sprint no	Schedule (start/end date) High level content
0	(08.09.2025 / 19.09.2025) - Finding out project/product requirements - Writing Project Plan document - Define product backlog for Sprint 1
1	(22.09.2025 / 03.10.2025) - Initialize project repository - Create initial CI/CD pipelines - Create architectural plans for mobile, web and backend (with priority on DB + backend) - Define a more detailed product backlog for the next x sprints
2	(06.10.2025 / 17.10.2025) - Finalize initial architecture at this point, so that multiple people can work on the same area - Define a prioritization order for features - Start developing first features – initial web and mobile app without of-line-first caching on mobile
3	(20.10.2025 / 31.10.2025) - Develop features according to the planned prioritization order

4	(03.11.2025 / 14.11.2025) - Develop features according to the planned prioritization order - Define and scope out what is feasible to do during the final implementation sprint
5	(17.11.2025 / 28.11.2025) - Implement the final features - Do not leave unfinished work that would be difficult for following groups to work on
6 (QA)	(01.12.2025 / 05.12.2025) - Quality assurance, testing with customer in staging environment, only changes are fixes to existing implementation

5.1.1 Iteration 0

During Sprint 0, the group familiarized themselves with the project topic. This included meetings and demonstration with the customer, as well as reading documents provided by the company that has collaborated customer. During this Sprint also the first version of the Project Plan document was written. The group is committed to continuing the project and wants to reimplement the product faithful to the North Star rewrite-plan. The first of three meetings was held with the Mindhive contact.

5.1.2 Iteration 1 (First Implementation Sprint)

The plan during sprint one is to implement at least the epic one, with partial completion of epics 2 and 3. These exist in Trello. Features in epic 1 include initialization of the applications and CI/CD in the repo. Epic 2 contains CI/CD expansion with quality gates and epic 3 is related to architectural planning. Effort is difficult to estimate, most likely all features of epics 2 and 3 cannot be implemented. The implementation of unfinished features will have to be continued in the next sprint. There are no real deliverables at this point, except potential architectural documentation. The tasks will be divided for team members accordingly; the aim is to have equal workload for everyone.

5.1.3 Iteration 2

During the first sprint, the initialization of the applications was nearly completed. An initial CI pipeline was created, and architectural planning for the mobile application began. At the end of the first week, the group responsible for Suunto watch integration was introduced, which required additional planning and coordination.

While this caused some minor architectural adjustments, the main impact was the redistribution and clarification of team responsibilities to better accommodate the new integration work. Consequently, no epics were fully completed in the first iteration, as the team concentrated on establishing the foundations for future development.

The focus for the second iteration will be to continue refining the architecture and begin active development of the web and mobile applications. Development will be based on the initial API specification, which is planned to be drafted and delivered during the first week of Iteration 2.

Since the backend is not yet available, mocks/stubs will temporarily replace it during implementation. This sprint will also mark the beginning of work on the first functional features of the application, focusing on map functionality and segments.

6 Requirements

6.1 Functional requirements (main goals)

As already mentioned in Chapter 3, the high-level goal of this project is to rewrite the existing Snowledge codebase to have a simpler and cleaner architecture. This means that the codebase has a single web frontend, mobile application, backend, and database. Rewriting the entire codebase involves a large amount of work. At the initial steps of the project, estimating the achieved degree of readiness in the end of the project is hard. This is why it has been agreed that the project group should focus on the quality of the implementation, rather than emphasizing the quantity of work produced.

The whole rewrite contains the following requirements:

- Initializing the codebase with the new structure.
- Building CI/CD pipeline to automate the workflow as much as possible.
- Creating single unified database, backend and web frontend.
- Writing a mobile application, designed to also function offline.

These would be implemented as time allows, focusing on providing a functioning state in the end of the project work. The architecture of the restructured codebase is a monorepo, i.e. a single repository houses all the different components. Reworking the CI/CD pipeline allows to automate e.g. testing, linting, deployment and use of strict quality gates to help produce as high quality code as possible. The rebuild mobile UI and web frontend would have the same visual features and functionality as the existing implementation.

Due to joining forces with the second project group in the late phases of Sprint 1, the objectives of this project group's work have been slightly adjusted. The second project group takes the responsibility of designing and implementing the backend according to the North Star document. Therefore, the remaining task for this project group is to design and implement the web and mobile frontends, following the description in the North Star vision. After the corresponding features of these project components, i.e. frontend and backend, have been implemented, they are integrated together to build a fully functional application.

6.2 Non-functional requirements goals

The following chapters discuss the different requirements and goals for the project from different aspects. In chapters 6.2.1–6.2.3 some metrics and acceptance levels have been mentioned. The plan is that these would be validated only, when necessary. In other words, they serve more as a guideline for development and planning. If no sufficient implementation exists in the end of the project, these metrics and acceptance levels will continue to live as a vision or target for the following project groups, guiding implementation to a high standard from the very beginning.

6.2.1 Usability goals

The user experience and usability of the web and mobile applications are important aspects to the customer. Intuitiveness and flow of information have been stated to be some features of good user experience. The customer didn't define any metrics to measure the completion of these goals, but the following table would serve as suggestion for a guideline for validating these goals in the future.

Metric	Acceptance Level
Task completion rate	≥ 80% of test users complete core workflows unaided
Error rate	≤ 5% errors in critical tasks
Time-on-task	Median < 2 minutes for core workflows
User satisfaction	Average score ≥ 4/5 in usability testing

6.2.2 Performance goals

The developed application stack must at least be able to handle the expected number of users during the winter season. This has been said to be few thousand users over the season. The customer didn't define any metrics to measure the completion of these goals, but the following table would serve as suggestion for a guideline for validating these goals in the future.

Metric	Acceptance Level
Response time	95% of requests complete in < 2 seconds
Concurrent users	≥ 200 concurrent sessions without > 70% CPU/memory use
Throughput	≥ 20 requests/second sustained under normal load

6.2.3 Reliability goals

Given that the application is expected to serve hundreds or potentially thousands of users, the system should be designed with reliability in mind, even though no goals have explicitly been set. Implicitly, this suggests general objectives such as high availability, scalability, resilience to failure and stable performance under the anticipated user load. The customer didn't define any metrics to measure the completion of these goals, but the following table would serve as suggestion for a guideline for validating these goals in the future.

Metric	Acceptance Level
Uptime / Availability	≥ 99% uptime during operating season

Mean time to recover	≤ 2 hours for critical incidents
Error rate	≤ 1% of all requests fail (e.g. HTTP 5xx responses, crashes)

6.2.4 Security goals

The following security baseline has been defined for the application stack:

- All secrets utilized by the CI/CD pipeline are stored and handled using appropriate manager. These secrets are also handled to Pallaksen Pöllöt.
- HTTPS should be utilized everywhere in the production environment. HSTS should be used in companion.
- Inputs should be validated, and outputs should be encoded. Error handling related to this should be centralized.
- Rate limiting and basic WAF should be applied.
- Users should be authenticated for any endpoint that changes data.
- The CI pipeline should scan for vulnerabilities and update dependencies.

In addition, the rules and regulations stated by GDPR should be adhered to in terms of security.

6.3 User interface requirements (main goals)

As already mentioned in Chapters 6.1 and 6.2.1, the main requirements for the user interfaces, being the web frontend and mobile UI, include the following:

- Share visual features and functionality with the existing implementation.
- Intuitiveness and flow of information should be emphasized to guarantee a high level of user experience and usability.
- Improvements to the existing user experience as deemed necessary.

6.4 Budget goals

The budget in this project refers to the working hours available for the duration of the project. These hours are allocated to different sections of the project as follows:

- Planning & Project Management 5%
- Implementation 40%
 - Web 40%
 - Mobile 55%
 - CI/CD 5%
- Testing 15%
- Documentation 15%
- Meetings 20%
- Mandatory Course Events 5%

7 Risk management

A key factor to a project's success is finding and reacting to any possible arising risk, before it gets to affect the schedule or progress of the project in a negative manner. The aim is also to reduce the likelihood for known risks to happen whenever possible.

The risks have many sources, but the most common ones include group members, the customer, software technology, and third parties, i.e. the overall project environment.

All risks are classified using a five-level scale, being Very Low / Low / Medium / High / Very High. Also, a standard number scale can be used, being 1-5. Each risk is described using fields Explanation, Severity / Impact, Probability, as well as Size / Importance.

During the project, an updated list of risks is maintained using the Metrics Monitoring Tool (MMT).

7.1 Risk list

The following Table 7.1 includes all the identified risks in the beginning of the project. A separate and updated list of risks is maintained using the Metrics Monitoring Tool (MMT).

Table 7.1. Project risks.

Risk ID	Explanation, Severity / Impact, Probability, Size / Importance
Illness / Absence	Explanation: Team member or one of the customer representatives gets sick or is absent during the project work. Severity: Medium (3), Probability: Low (2), Importance: Medium (3)
Lack of Communication	Explanation: Not enough communication between the group members or between the group and customer representatives, which affects the resulting product or its quality. Severity: High (4), Probability: Medium (3), Importance: High (4)
Slow Communication	Explanation: The group members or the group and the customer representatives are communicating, but responding to questions or previous message takes a long time. Severity: High (4), Probability: Medium (3), Importance: High (4)
Reduced Code Quality Due to Inexperience	Explanation: If developers are inexperienced with new technologies, then mistakes and inefficient coding practices may occur, leading to rework, defects, and reduced overall code quality. Severity: Very High (5), Probability: Low (2), Importance: Very High (5)
Integration Challenges	Explanation: If the new tools and technologies are difficult to integrate with existing systems or workflows, then unexpected technical issues may arise, leading to milestone delays and additional design or architectural changes. Severity: Very High (5), Probability: Medium (3), Importance: Very High (5)
Delays Due to Sudden Changes in Requirements	Explanation: If project requirements are changed or added unexpectedly during development, then the team will need to re-plan, rework completed tasks, or adjust designs, leading to schedule slippage and potential cost overruns. Severity: Very High (5), Probability: Low (2), Importance: Very High (5)

7.2 Risk monitoring

The risks are monitored during the Internal Weekly Meetings. They are managed using the Metrics Monitoring Tool (MMT). Monitoring them is a common task of all group members. Monitoring risks includes analysing for new risks and the realisation of known and existing risks.

8 References

- [1] Pallaksen Pöllöt. Available at: <https://www.pallaksenpollot.com/> (Accessed 17 September 2025)
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9 Open issues

None.

10 Ideas for further development

None at the time of writing.

APPENDIX A: Mindhive Current State Document

APPENDIX B: Mindhive North Star Document